AFIN3029 - Sons of Midas Assignment

Question 1: Understanding the forward and option contracts

a) **Calculate the per annum 6-month Gold lease rate that is implied from the 6-month forward contract on Gold with a forward price of USD 2,202.64.**

In the model: $F_0 = S_0 e^{(r-q)T}$, q (the implied Gold lease rate) can be treated as the unknown variable of interest, such that: $2202.64 = 2157.41e^{(0.0565-q)0.5}$. To re-arrange for q we have:

$$2202.64 = 2157.41e^{(0.0565-q)0.5}$$

$$\frac{2202.64}{2157.41} = e^{(0.0565-q)0.5}$$

$$In(\frac{2202.64}{2157.41}) = In(e^{(0.0565-q)0.5}) \Rightarrow In(\frac{2202.64}{2157.41}) = (0.0565 - q)0.5$$

$$\frac{In(\frac{2202.64}{2157.41})}{0.5} = (0.0565 - q) \Rightarrow q = 0.0565 - \frac{In(\frac{2202.64}{2157.41})}{0.5} = 0.01500357444..$$

Throughout the rest of the assignment, I used the solver value for q as it was more precise and convenient. By first selecting a dummy value for q, say 2.5%, we can input the value of F_0 to be given by:

$= 2157.41 * EXP((0.0565 - 0.025) * 0.5)$ in Excel, we get: $q = 1.50036...\%, \ using\ solver.$

b) **Explain what the Gold lease rate represents and what it means when it is negative, which it has been in recent times.**

As Gold is an investment asset but can be used for industrial operations, the users of Gold can experience the physical benefit of the commodity not obtained by holders of forward contracts. Holding physical Gold will allow a firm to continue its manufacturing processes if Gold is a required input, whereas holding a forward contract on Gold cannot be used for such a purpose. Therefore, the benefit of holding Gold physically is the difference between the convenience yield and the storage costs of Gold: $Gold\ lease\ rate = convenience\ yield - storage\ costs \Leftrightarrow q = y - u$, which represents the cost incurred when borrowing Gold or the income received when lending out Gold. Since the borrowing and lending are unsecured, q could also be regarded as a risk premium for the possibility of the Gold not being returned, less the storage costs (Le, A. and Zhu, H., 2013).

From this, we can represent the forward price as $F_0 = S_0 e^{(r-(y-u))T}$, which given $q = y - u$ reduces to our familiar expression of $F_0 = S_0^{(r-q)T}$ (Hull et al. 2013). What this therefore means if q is negative ($q < 0$), then the costs for storing Gold (u) are more than the benefits (y), or more formally, $y - u < 0$. Moreover, if q is negative, then we have $F_0 = S_0^{(r+q)T} > F_0 = S_0^{(r-q)T}$, so when the Gold lease rate is negative, it implies the expectation of higher forward prices, and therefore higher future spot prices, under the assumption of no arbitrage (and therefore no basis at time T). This would imply that owners of Gold are experiencing large storage costs, recieving a negative return when paying others to borrow the Gold from them. The negative Gold yield can indicate a high demand for physical Gold when investors prefer tangible assets during recent times of high inflation (above the RBA's 2-3% target band), with inflation placing upward pressures on storage costs.

c) **When pricing a forward contract, it is often important to consider a replicating portfolio – a portfolio that replicates the forward contract's payoff. Demonstrate how you would construct a portfolio of the underlying asset and cash to replicate a 6-month forward contract to buy 1 ounce of gold at a forward price of USD 2,202.64. Your answer should show the cash flows at t = 0 and payoff at t = 6 months for each transaction. Note that your answer must not include option contracts**

The forward contract to buy one ounce of Gold at a forward price of USD 2,202.64 in 6 months can be replicated by purchasing Gold spot, borrowing USD to finance and lending out Gold. This would replicate the net cash flows at 0 months and 6 months for the forward contract. Specifically, the net cash flow at time zero should be zero, and the net cash flow at 6 months should be -2202.64, with an inflow of one ounce of Gold.

If we were to buy exactly one ounce of Gold at the current spot price, lend it out and then keep it for 6 months, using the lease rate earned to reduce borrowing costs, then the net cash flow should be given by $\Sigma_{6\ months\ time} = -(S_0 \times e^{rT}) + (S_0 \times e^{qT} - S_0) = F_0$. However, this value would be greater than $-\ 2202.64$ in absolute terms. So, when purchasing 1 ounce at time zero, by borrowing 2157.41, the equivalent q income would be $(S_0 \times e^{qT} - S_0) = 2157.41e^{0.0150036\times0.5} - 2157.41 = 16.2453...$ So the total cost for one ounce would be: $2157.41 \times e^{0.0565\times0.5} - 16.2453 = 2202.980571 \neq 2202.64$. Therefore, we should consider purchasing less than an ounce of Gold.

Since the present value of one ounce of Gold in terms of Gold in the future is $1e^{-qT}$, we should purchase $1e^{-1.50036...\%\times0.5} = 0.9925$ ounces of Gold $(rounded\ to\ 4\ d.p.)$. This is equivalent to borrowing the present value of the forward price discounted at the risk-free rate to purchase the 0.9225 ounces of Gold at time 0: $F_{-6\ months} = 2202.64 \times e^{-0.0565\times0.5} = 2141.2861 = 0.9925 \times 2157.41.$

Therefore, the replicating portfolio would involve:

1. $At\ time\ t = 0,\ borrow\ S_0 = 2202.64 \times e^{-0.0565\times0.5} = 2141.2861$
2. $At\ time\ t = 0,\ buy\ 0.9925\ ounces\ of\ Gold\ for\ -\ 2141.2861\ and\ lease\ it\ out\ at\ q$
3. $At\ time\ t = 6\ months,\ collect\ Gold\ from\ borrower = 0.9925$
4. $At\ time\ t = 6\ months,\ collect\ the\ lease\ payment\ on\ Gold\ from\ the\ borrower$, which is equivalent to $0.0075\ ounces\ of\ Gold\ (e^{0.0150036..\times0.5} - 1 = 0.0075..)\ or$ $0.0075.... \times 2202.64 = \$16.58587065\ (assuming\ no\ basis\ at\ t = 6\ months)$.
5. $At\ time\ t = 6\ months,\ repay\ USD\ borrowings = 2141.2861 \times e^{0.0565\times0.5} = 2202.64$

Tabulating this gives:

t= now			t= 6 months		
Transaction	Gold	USD	**Transaction**	**Gold**	**USD**
Borrow USD		$ 2,141.29	Repay USD		-$2,202.64
Buy Gold	0.9925	-$ 2,141.29	Keep Gold		
Lend Gold	-0.9925		Receive Gold	0.9925	
			Receive +q	0.0075	
Net CF	0	0	**Net CF**	1	-$2,202.64

d) Verify that put-call parity holds for all the option contracts listed above. Present your results in a table.

For European options, given identical strike prices and maturity, a portfolio with a call option (c) and Xe^{-rT}(PV of the strike price) will be equal to the portfolio of a put option (p) and the present value of the option's underlying asset (S_0e^{-qT}). This assumes no-arbitrage as each position's costs and resulting payoffs should be the same: $c + Xe^{-rT} = p + S_0e^{-qT}$. To test this equality, we will apply this formula's LHS in one column and the RHS in another. We can then use an if statement (=IF(LHS_column=RHS_column,"True","False")) to automatically tell us if the condition in the formula is true.

Please also note that fixed values in the formulas that are the same for all $c + Xe^{-rT}$ and $p + S_0e^{-qT}$ columns are:

- $q = 1.50036..\%$, $r = 5.565\%$, $S_0 = 2157.41$, $T = 0.5$

Strike price (USD per ounce)	OTC European call option premium (USD/ounce)	OTC Euro put option premium (USD/ounce)	C+Xe-rT	P+S*e-(q)*T	Parity?
$ 2,480.00	$ 15.94	$ 285.57	$ 2,426.86	$ 2,426.86	False
$ 2,375.00	$ 33.09	$ 200.64	$ 2,341.94	$ 2,341.93	False
$ 2,265.00	$ 64.70	$ 125.32	$ 2,266.61	$ 2,266.61	False
$ 2,202.64	$ 90.57	$ 90.57	$ 2,231.86	$ 2,231.86	False
$ 2,157.41	$ 113.33	$ 69.36	$ 2,210.65	$ 2,210.65	False
$ 2,050.00	$ 180.90	$ 32.51	$ 2,173.80	$ 2,173.80	False
$ 1,940.00	$ 267.35	$ 12.02	$ 2,153.31	$ 2,153.31	False
$ 1,835.00	$ 360.99	$ 3.59	$ 2,144.88	$ 2,144.88	False

Initially, we note that the parity condition according to the if statement column does not hold for any option contract. However, given the estimation error in our value of q from part a, and the rounded values provided for; the strike price, premiums and current Gold price, we can assume that the put-call parity does hold. This assumption is justifiable since the parity is a no-arbitrage condition, where commissions and other trading fees would be a limit to arbitrage on small price differentials, such as in the case of values found in the table. This assumes there was no measurement error in the values provided (spot price, forward price etc).

To determine how many decimal places contain this error, we can add another column: Difference, which just subtracts the $c + Xe^{-rT}$ columns by $p + S_0e^{-qT}$:

Strike price (USD per ounce)	OTC European call option premium (USD/ounce)	OTC Euro put option premium (USD/ounce)	C+Xe-rT	P+S*e-(q)*T	Parity?	Dif
$ 2,480.00	$ 15.94	$ 285.57	$ 2,426.86	$ 2,426.86	False	$ 0.00
$ 2,375.00	$ 33.09	$ 200.64	$ 2,341.94	$ 2,341.93	False	$ 0.01
$ 2,265.00	$ 64.70	$ 125.32	$ 2,266.61	$ 2,266.61	False	$ 0.00
$ 2,202.64	$ 90.57	$ 90.57	$ 2,231.86	$ 2,231.86	False	$ 0.00
$ 2,157.41	$ 113.33	$ 69.36	$ 2,210.65	$ 2,210.65	False	-$ 0.00
$ 2,050.00	$ 180.90	$ 32.51	$ 2,173.80	$ 2,173.80	False	$ 0.00
$ 1,940.00	$ 267.35	$ 12.02	$ 2,153.31	$ 2,153.31	False	$ 0.01
$ 1,835.00	$ 360.99	$ 3.59	$ 2,144.88	$ 2,144.88	False	$ 0.00

Looking at the differences between contracts and their respective parity values in columns, we can see that when rounded to 1.d.p, the differences disappear. We can therefore apply =ROUND(value,1) to the formulas in columns $c + Xe^{-rT}$ and $p + S_0e^{-qT}$:

Strike price (USD per ounce)	OTC European call option premium (USD/ounce)	OTC Euro put option premium (USD/ounce)	C+Xe-rT	P+S*e-(q)*T	Parity?	Dif
$ 2,480.00	$ 15.94	$ 285.57	$ 2,426.90	$ 2,426.90	True	$ -
$ 2,375.00	$ 33.09	$ 200.64	$ 2,341.90	$ 2,341.90	True	$ -
$ 2,265.00	$ 64.70	$ 125.32	$ 2,266.60	$ 2,266.60	True	$ -
$ 2,202.64	$ 90.57	$ 90.57	$ 2,231.90	$ 2,231.90	True	$ -
$ 2,157.41	$ 113.33	$ 69.36	$ 2,210.60	$ 2,210.60	True	$ -
$ 2,050.00	$ 180.90	$ 32.51	$ 2,173.80	$ 2,173.80	True	$ -
$ 1,940.00	$ 267.35	$ 12.02	$ 2,153.30	$ 2,153.30	True	$ -
$ 1,835.00	$ 360.99	$ 3.59	$ 2,144.90	$ 2,144.90	True	$ -

Therefore, the mathematical condition of put-call parity holds when considering computational, rounding and arbitrage limits. Empirical research from Shleifer and Vishny (1997) substantiates this conclusion, finding that seemingly profitable arbitrage opportunities are unfeasible or too risky, particularly when the price discrepancies are small. This would imply that even if there were no rounding or computational errors, the fact that the differences in the prices are so small makes the put-call parity condition effectively hold.

e) **Why are the European call and put premiums the same when the strike price is USD 2,202.64? Explain your answer.**

Given the parity condition: $c + Xe^{-rT} = p + S_0e^{-qT}$:

$c = p + S_0e^{-qT} - Xe^{-rT}$

$\therefore If: S_0e^{-qT} - Xe^{-rT} = 0\ ,\ then: c = p$

This parity relies on the benefit of exercising the call ($buying\ at\ X\ when\ S_T > X$) being equivalent to exercising the put option ($selling\ at\ X\ when\ S_T < X$). Parity therefore occurs when the expected future spot price of Gold (adjusted for its yield and TVM) matches the present value of the strike price. With respect to a portfolio that replicates a call option using a put, zero coupon bond, and Gold ($c = p + S_0e^{-qT} - Xe^{-rT}$), or a put replicated using a call, zero coupon bond, and Gold ($p = c + Xe^{-rT} - S_0e^{-qT}$), this indicates that the market perceives the risks and benefits of both strategies (owning the call versus owning the put) as equal. Therefore, for the 2202.64 strike, we can prove that $S_0e^{-qT} - Xe^{-rT} = 0$ given:

$2157.41e^{-1.500458...\%\times 0.5} - 2202.64e^{-5.565\%\times 0.5} = 2141.286124 - 2141.286124 = 0$

$\therefore c = 90.57 + 0 \Rightarrow 90.57 = 90.57$ **,** such that: $c = p$.

This is clearly not the case for other contracts, such as a strike price of USD $2480:

$S_0e^{-qT} = 2141.286124$ and $Xe^{-rT} = 2410.920344$. Hence $15.94 \neq 285.57 \Rightarrow c \neq p$

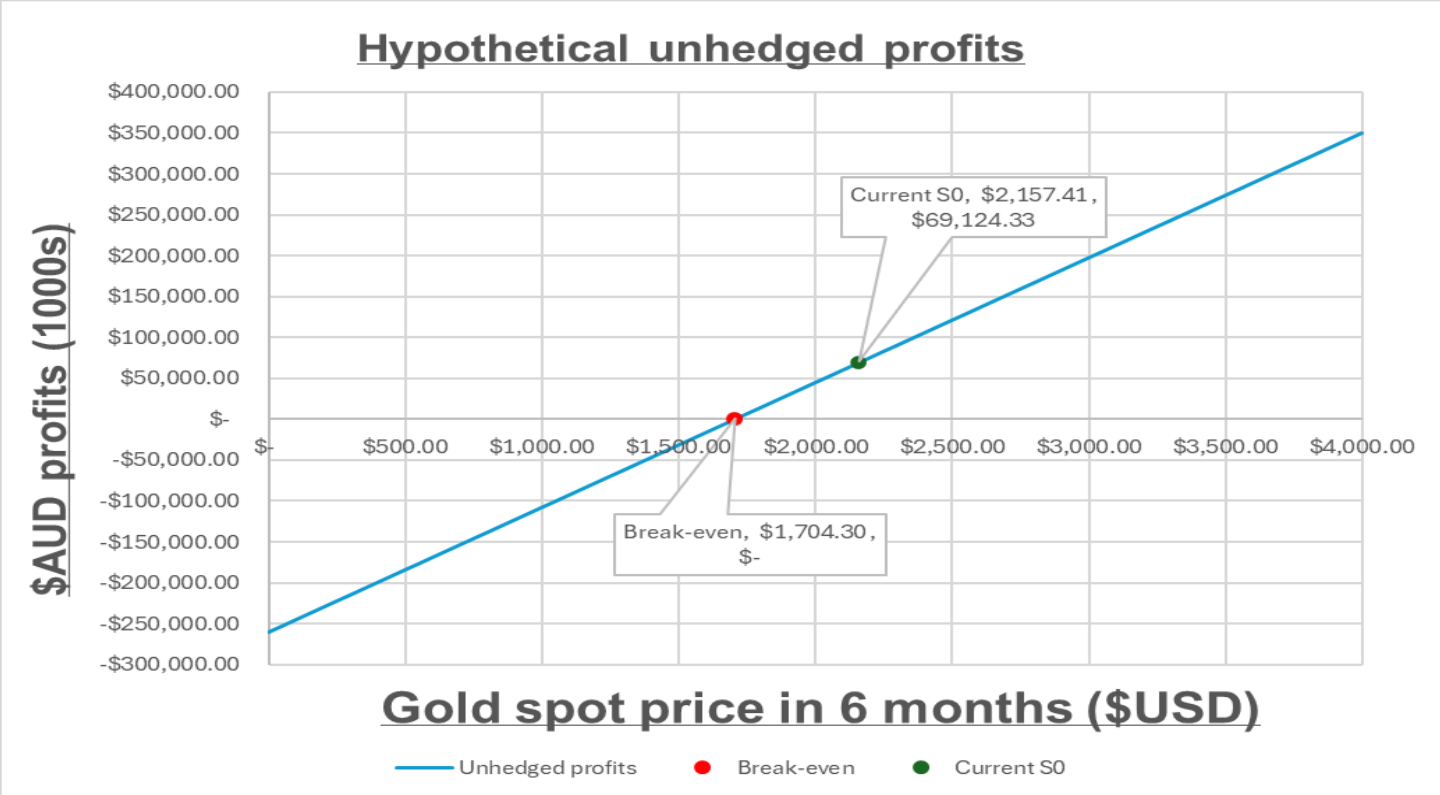
<u>Question 2: Hedging strategies (45 marks)</u>

a) **Produce a profit diagram in AUD for the unhedged profits of Sons of Midas. Your profit diagram should have profit in AUD on the y-axis and ST on the x-axis. Prepare your workings in Excel and include the profit diagram in your assignment submission. What movements in the Gold price hurt/benefit Sons of Midas? (5 marks)**

Given that Midas has a long position in Gold, their unhedged profits, in AUD, are based on the net of the AUD sale price and AUD costs. For each hypothetical Gold spot price in 6 months, we can graph the potential profits Midas could experience. We start by formulating the profit function in a table form for each spot price:

Hypothetical $S_{0.5}$ $(\$USD/ounce)$	**Hypothetical revenue** $(\$AUD/ounce)$	**Costs** $(\$AUD/ounce)$	**Profit** $(\$AUD)$
0	**To convert USD sales revenue into AUD:** $S_{0.5}(\$USD) \times \frac{1}{0.6555} = S_{0.5}(\$AUD)$	**Are assumed to be a constant** $(\$2600)$	$\Pi = (Revenue(\$AUD) - Costs(\$AUD)) \times 100k$
0+50	**76.28**	**2600**	**-252.372 million**
……	……	……	……
4000	**6102.21**	**2600**	**+350.221 million**

Diagrammatically, this results in:



Therefore, any % change in the spot price that is $\leq -\ 21.00249\%\ (\frac{1704.30}{2157.41} - 1)$ will be harmful to Midas, as this is past the price where they break even ($S_T < 1704.30(USD)$). So long as the change is greater than this, given costs are constant, Midas will make a profit. Therefore, movements below $2157.41 (USD) but greater than $1704.30 (USD) will be less desirable but still result in Midas achieving a profit per ounce on its sale of Gold. According to the graph, anything above $2157.41(USD) will benefit Midas even more, as profits will increase linearly as a function of the net of the spot price in 6 months and the cost of producing Gold.

b) Explain how you could hedge the Gold exposure of Sons of Midas with the 6- month forward contract at a forward price of USD 2,202.64. Produce a profit diagram in AUD showing the unhedged profits and hedged profits. (8 marks)

As Son's of Midas is inherently long Gold, it faces the risk that the price of Gold will decrease and that the price at which it can sell the Gold it produces will be lower than its production costs. Midas can hedge this downside price risk by locking in the sale price of Gold using a short position in the 6-month forward contract at a forward price of USD 2,202.64. Specifically, this will involve 0 net transaction costs at time 0 and entering into the short forward agreement. In 6 months, they will honour the forward contract and sell each ounce of Gold for $F_0 = \ \$2,202.64\ (USD)$. This forward contract is assumed to be cash settled, or if it were physically delivered, it would need to be closed out in 6 months.

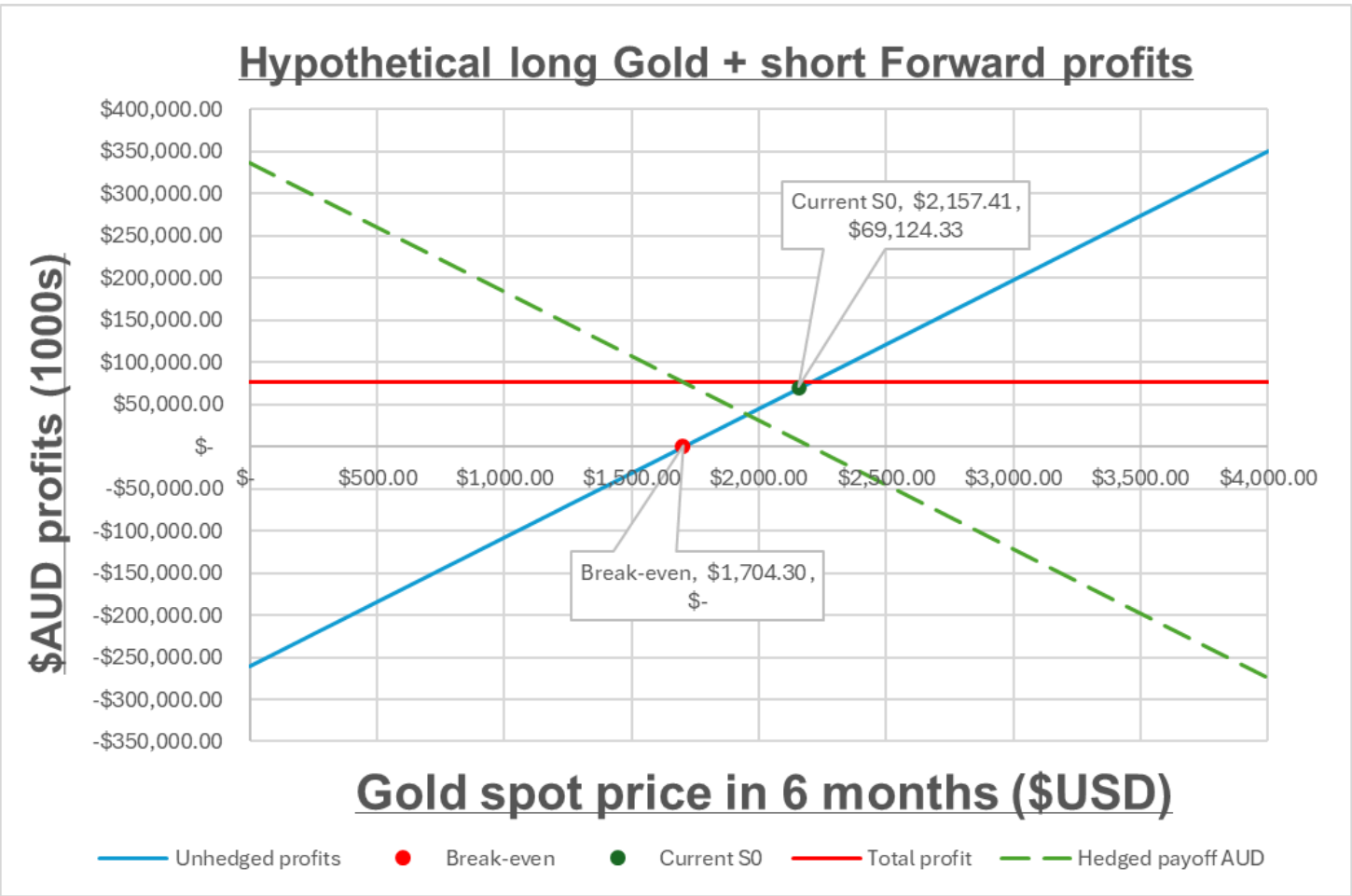
Since the total profit for the hedging strategy is the profit from the unhedged position (long) in Gold and the profit from the hedge (short forward agreement), we calculate the intrinsic payoff for the forward as $F_0 - S_T$. We repeat this for each S_T from 0 to 4000. We then sum each hedged profit value for a given S_T with unhedged profit $(S_T - 2600)$ to find the total profit from the long position in Gold combined with the short forward position:

$\Pi^{AUD}_{Total} = Unhedged\ profit\ +\ hedged\ profit = [S_T - 2600] + [F_0 - S_T]$, which simplifies to

$\Pi^{AUD}_{Total} = F_0 - 2600$ without considering any time value or additional costs such as commission. Tabulating results gives:

Hypothetical spot price in 6 months (USD)	Hypothetical spot price in 6 months (AUD)	AUD production costs	Unhedged profits (AUD)	Forward price (AUD)	Hedged payoff (AUD)	Total hedged profit (AUD)
$S_{0(USD)} = 0$	$S_{0(USD)}/0.6555$	$C_0 = 2600$	$(S_{0(AUD)} - C_0) \times 100k$	$F_{0(USD)}/0.6555$	$(F_{0_{(AUD)}} - S_{0(AUD)}) \times 100k$	$(F_{0(AUD)} - 2600) \times 100k$
$S_{0(USD)} = 0 + 50$	$50/0.6555 = 76.28$	$C_0 = 2600$	-252.372 million	3360.24	328.397 million	76.024 million
……	……	……	……	……	……	……
$S_{0(USD)} = 4000$	$S_{0(AUD)} = 6102.21$	$C_0 = 2600$	$+350.221$ million	3360.24	-274.197 million	76.024 million

Diagrammatically, since the F_0 is a constant here, we get a profit that is also constant for all S_T as $\$760.24 AUD/ounce$, or $76.024 million(AUD) (given the exposure of 100,000 ounces):



From the graph, we can note a significant risk-return tradeoff represented by the opportunity cost of entering the forward contract instead of remaining unhedged. This is reflected in the decrease in the hedged payoffs as the spot price of Gold increases, whilst the unhedged profit increases as the price of Gold rises. Therefore, when the spot price of Gold is above the forward contract price, the hedged payoff becomes negative, representing a strengthening of the basis. Assuming no counterparty credit risk, this constant total profit structure results in a limited upper bound on Midas's total profits.

c) From the option contracts listed above, explain how you could hedge Sons of Midas's Gold exposure with a put option. Which put option have you chosen, and why? Produce a profit diagram in AUD showing the unhedged profits and hedged profits resulting from your chosen put option. (12 marks)

As Son's of Midas is inherently long Gold, it faces the risk that the price of the Gold it sells will decrease below its production costs. This means it can hedge this downside price risk by purchasing a put option to lock in the sale price of Gold. To evaluate each option based on the downside risk it protects, we can consider the differences in profitability across each put option based on whether it will be exercised and, thus, if it is a net benefit to be exercised or not. This is because Midas would sell at the strike price in 6 months whilst incurring an upfront expense on the option at the size of the premium paid.

The unhedged profits are simply $S_T - c_{production}$, however, the hedged profits are $Payoff - premium = max(K - S_T, 0) - p$. Here, the total profit for the long Gold + long put hedge strategy would, therefore, be given by a "married put" profit function:

- $If\ max(K - S_T, 0) > 0$: $Profit = K - c_{production} - p$ such that the put will be exercised, where $K = strike\ price$, $p = premium\ paid\ on\ put$, $c_{production} = production\ costs\ of\ Gold\ for\ Midas$.
- Contrarily, $if\ max(K - S_T, 0) = 0$: $Profit = S_T - c_{production} - p$ such that the put purchased will not be exercised, where $S_T = price\ of\ Gold\ in\ 6\ months$.
    - Note also that $c_{production}$ is a proxy for S_T in standard long asset + long put calculations.

Scaling this for the sale of 100,000 ounces (here represented in 1000s units for graphing), we can tabulate each of these profits for the different strikes and premiums of each option based on their payoffs for a given future price of Gold:

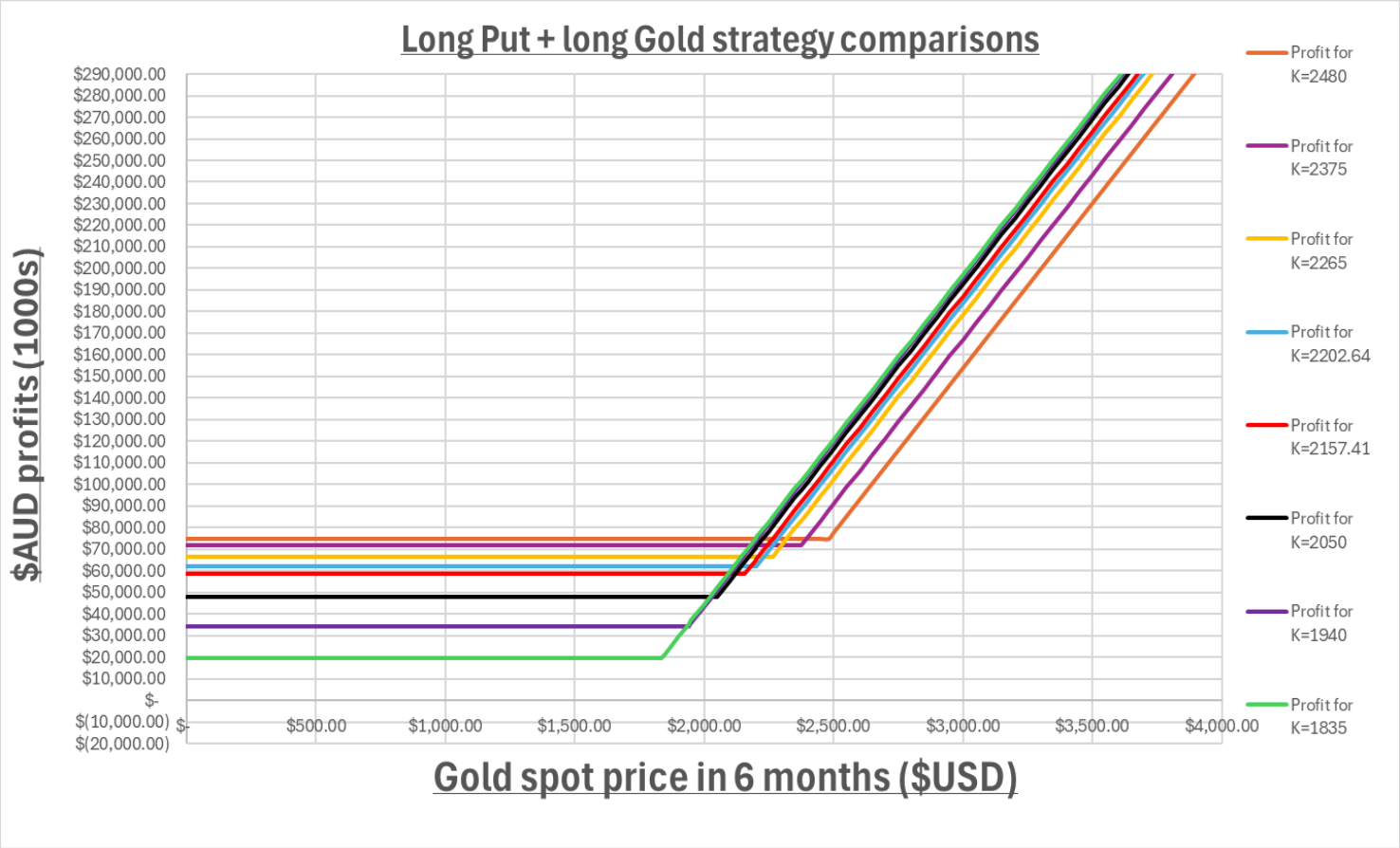
ACCRINT =(IF([@[Payoff for K= 2480.00]]>0,$D$2-$F$2-$B$7,[@[Hypothetical S_0.5 (USD)]]-$F$2-$B$7)/$B$5)*100

IF(logical_test, [value_if_true], [value_if_false])

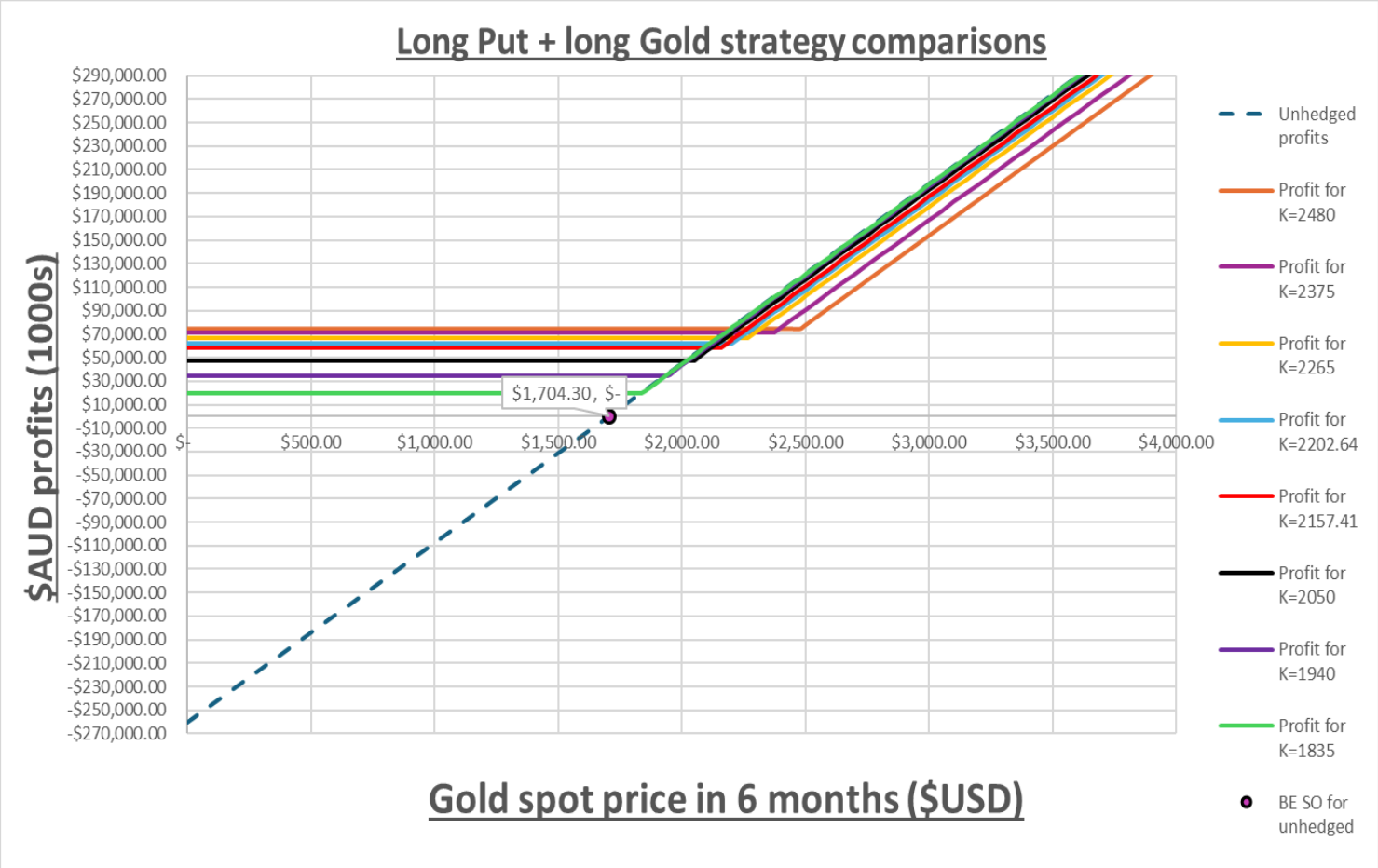
	A		C	D	E	F	G
1				Strike price (USD per ounce)	OTC European call option premium (USD/ounce)	OTC Euro put option premium (USD/ounce)	
2	Gold spot price(USD/ounce)	$ 2,157.41		$ 2,480.00	$ 15.94	$ 285.57	
3	Gold forward price for delivery in 6 months (USD per ounce)	$ 2,202.64		$ 2,375.00	$ 33.09	$ 200.64	
4	6-month US interest rate (c.c.)	5.65%		$ 2,265.00	$ 64.70	$ 125.32	
5	Exchange rate (XUSD/1AUD)	$ 0.6555		$ 2,202.64	$ 90.57	$ 90.57	
6	Costs(fixed per ounce) in AUD	2600		$ 2,157.41	$ 113.33	$ 69.36	
7	Costs USD	$ 1,704.30		$ 2,050.00	$ 180.90	$ 32.51	
8	q?=	1.50036%		$ 1,940.00	$ 267.35	$ 12.02	
9	Will sell (in 6 months)	100,000		$ 1,835.00	$ 360.99	$ 3.59	
10							
11							
12	Hypothetical S_0.5 (USD)	Payoff for K= 2480.00	Profit for K=2480	Payoff for K= 2375	Profit for K=2375	Payoff for K= 2265	Profit for K=2265
13	$ -	$ 3,783.37	K= 2480.00]]>0,	$ 3,623.19	$ 71,710.1449	$ 3,455.38	$ 66,419.5271
14	$ 50.00	$ 3,707.09	$ 74,771.9298	$ 3,546.91	$ 71,710.1449	$ 3,379.10	$ 66,419.5271
15	$ 100.00	$ 3,630.82	$ 74,771.9298	$ 3,470.63	$ 71,710.1449	$ 3,302.82	$ 66,419.5271
16	$ 150.00	$ 3,554.54	$ 74,771.9298	$ 3,394.36	$ 71,710.1449	$ 3,226.54	$ 66,419.5271
17	$ 200.00	$ 3,478.26	$ 74,771.9298	$ 3,318.08	$ 71,710.1449	$ 3,150.27	$ 66,419.5271
18	$ 250.00	$ 3,401.98	$ 74,771.9298	$ 3,241.80	$ 71,710.1449	$ 3,073.99	$ 66,419.5271

Chart Area

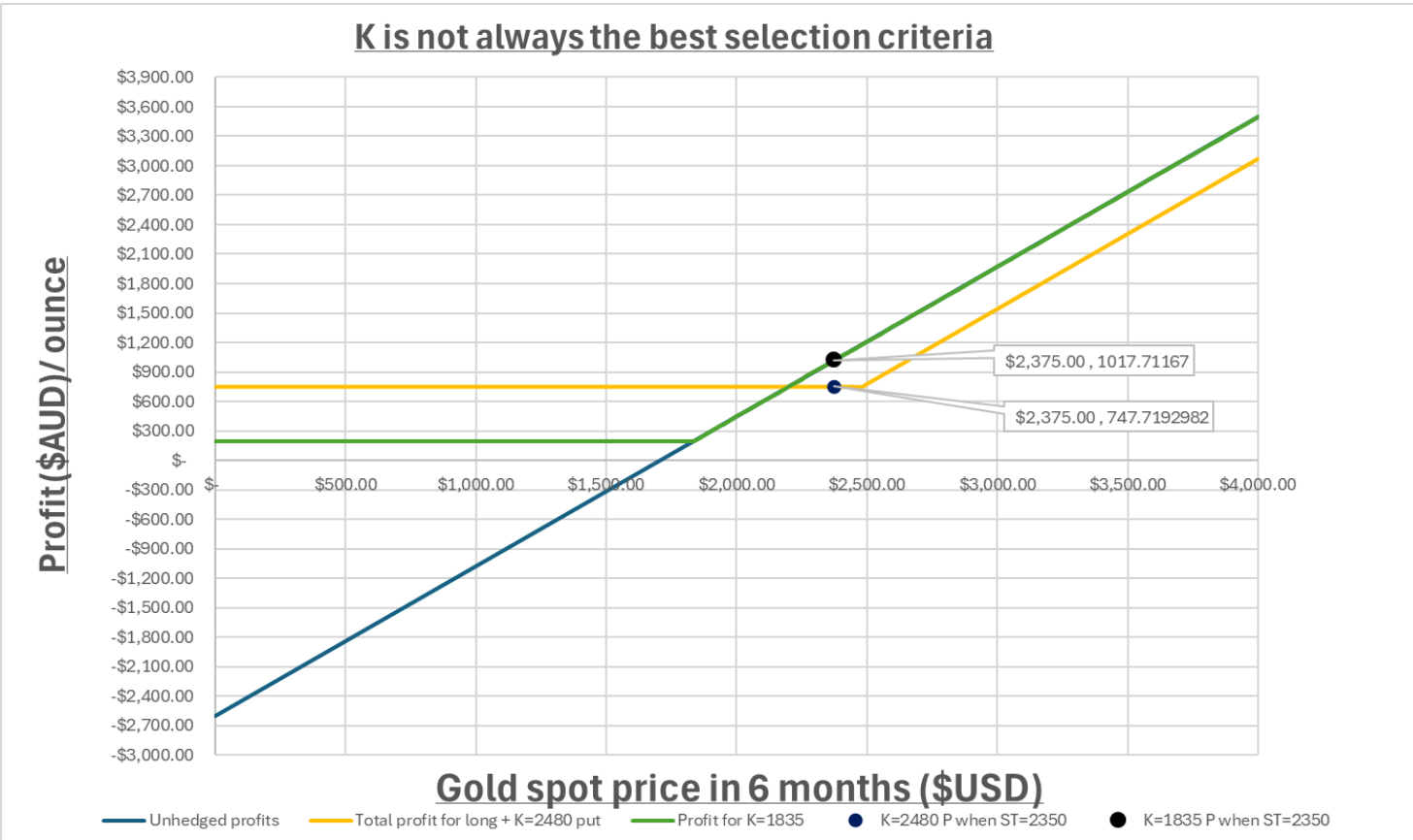
By plotting the total hedged strategy for each put option, we are given the following profit diagram that compares all available options:



When comparing this with the unhedged exposure, we also get the following graph:



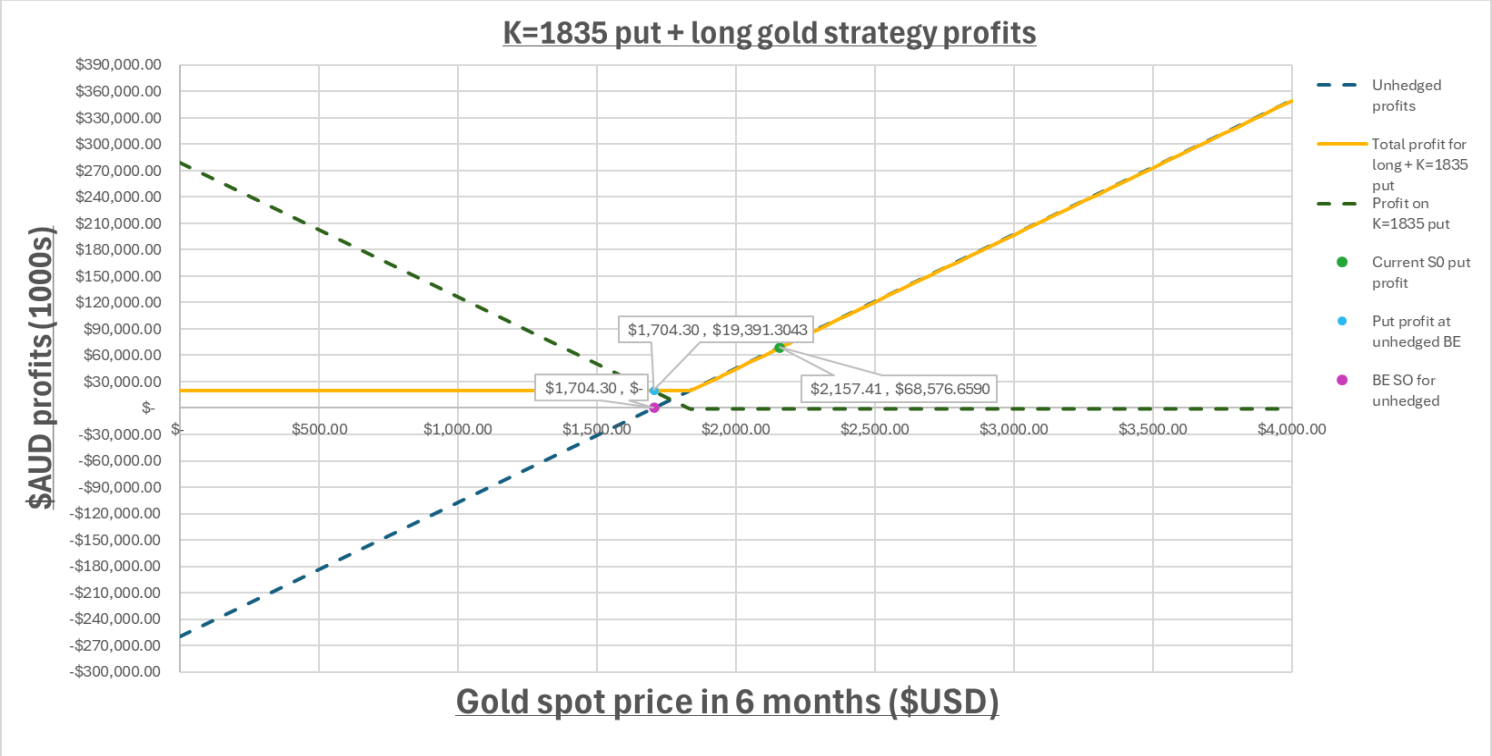
Since the AUD costs are $production\ cost\ = \$2600AUD/ounce$ and $premium\ = \frac{3.59(USD)}{0.6555} = \$5.477AUD/ounce$, at the lowest available sale (strike) price of $\frac{1835(USD)}{0.6555} = \$2799.39AUD/ounce$, the benefits of each put option clearly outweigh their costs, assuming they expire ITM ($\$2799.39AUD > \$2605.477AUD$). Furthermore, a higher strike does not determine if a put option hedge is more desirable than another due to the different turning points of each strategy's profits. Using a per-ounce representation of profits to simplify the point, we can show this diagrammatically through the profit function for K=2480, the highest strike, versus K=1835, the lowest strike:



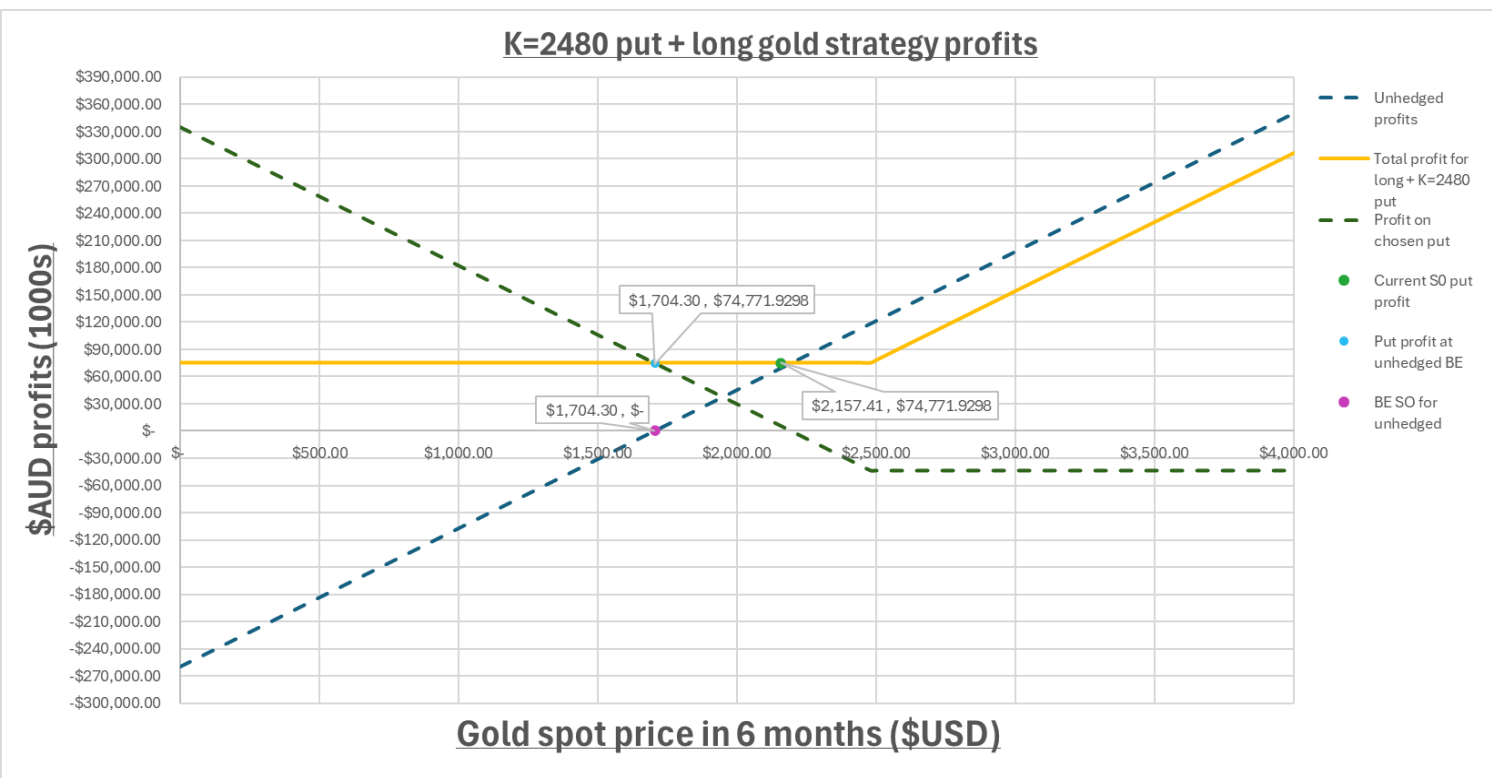
Moreover, the put with the strike price $K\ = \$2480(USD)$ is more expensive than all other available forms of hedging, at a total upfront cost of: $\left(\frac{-285.57\times100k}{0.6555}\right) = \$43.5652million\ (AUD)$. This could create substantial short-term cash flow issues for Midas, potentially requiring financing. We should instead consider a risk-to-return tradeoff defined by: $\frac{Average(hedged\ strategy\ profits - unhedged\ profits)}{premium}$. This allows us to compare the expected benefits the strategy provides above the benefits from remaining unhedged against the put's substantial upfront costs (and risk). This risk criteria provides a basis for analysing the hedging activity's utility, as hedging should seek to reduce Midas's cash flow uncertainty, not to increase profits.

Desc.	K=2480	K=2375	K=2265	K=2202.64	K=2157.41	K=2050	K=1940	K=1835
Avg benefit	$ 68,932.58	$ 71,801.59	$73,315.97	$73,246.98	$ 72,756.76	$ 69,982.28	$ 64,931.86	$ 58,975.81
Premium	$ 43,565.22	$ 30,608.70	$19,118.23	$13,816.93	$ 10,581.24	$ 4,959.57	$ 1,833.71	$ 547.67
Ratio	$ 1.58	$ 2.35	$ 3.83	$ 5.30	$ 6.88	$ 14.11	$ 35.41	$ 107.68

We find that the 1835 strike is preferred for each put, with a risk-to-return criterion of $\$107.68$ benefit above the profit realised when remaining unhedged per $1 of risk. This would enable Midas to participate in more of the increase in gold prices compared to the 2480 strike (since profit becomes $S_T - p - c_{production}$ when $S_T > K$) while also obtaining downside protection at a lower upfront cost of $\$0.547673million\ (AUD)$ ($\left(\frac{3.59}{0.6555}\right) \times\ 100,000$):



However, if Midas is more concerned about its downside risk in 6 months, the 2480 strike is preferred, as this would provide the highest minimum profit (lower bound) of $ 74.772 million(AUD) $\left(\frac{(2480-285.57-1704.30)\times 100{,}000}{0.6555}\right)$. This means that when there is no risk of default on the puts, Midas will always experience a positive profit, as it will always sell Gold at a price above its production costs. Therefore, if Midas has no financing issues for the purchase of the put and prefers the highest minimum profit, the put option used for hedging should indeed be the $K = \$2480(USD)$ strike:



The benefit of the higher strike is that if $S_T < K$, Midas will obtain a comparative advantage over unhedged competitors in short-term revenue, assuming similar costs. This could allow Midas to capture more market share, with stable cash flows increasing creditworthiness, allowing Midas to access more favourable borrowing terms for future capital investments or expansions. There is also no upper bound for this strategy, providing Midas with an unlimited profit upside when $S_T > K$.

d) From the option contracts listed above, explain how you could hedge Sons of Midas's Gold exposure with a collar strategy. Which option contracts have you chosen, and why? Produce a profit diagram in AUD showing the unhedged profits and hedged profits resulting from your chosen option contracts. (15 marks)

The construction of a collar will require:

1. Long Gold, purchasing for S_0 *(Midas inherently is long Gold, with constant production costs being $2600 AUD).*
   Therefore, $S_0 = \$2600(AUD)$
2. Long a put on Gold with strike price K_1
3. Short a call on Gold with strike price K_2
4. Ensure that $K_2 > K_1$, with both options having the same expiration date.

Thus we would need to consider the net cost of the collar and the tradeoff in minimising this when selecting the call option, which provides a limit on hedged profits, as well as the put option, as this will determine the limit on the downside of the hedging strategy. Puts with lower premiums will reduce this net cost; however, they will be more OTM, thus offering less downside risk protection.

By analysing just three combinations, the ones with the highest net premium received, the most neutral net premiums received, and the smallest net premium received (highest paid) (where the net premium on the collar strategy is $c - p$), we can establish a range of risk-return tradeoffs. Specifically, the highest net premium received would provide Midas with a substantial upfront cash flow whilst accepting the risk of a small upper bound in its profits. The neutral premium received would represent a minimisation of initial cash flows whilst maximising exposure to linear payoffs, increasing the difference between the lower and upper bounds of profits. The collar with the highest premium paid will also trade short-term cash flows and upper bound price exposure for significant downside protection (represented by its lower bound), effectively addressing the concerns of potential increases in Gold price volatility.

Including ITM puts in our analysis can also increase immediate downside protection, as they start with intrinsic value, protecting against declines below their strike price. Including ITM calls would also result in higher premiums, offsetting the cost of the puts, but at the expense of creating an upper bound on hypothetical profits closer to the current Gold price (El-Hassan et al. 2021).

Given that the net premium is $c - p$ and that $K_{call} > K_{put}$, an if statement can be used when combining each call and put strike in a matrix to calculate the net premiums (in USD/ounce) automatically. Starting from the cell that contains $K_{put} = 2480$ and $K_{call} = 2480$ to complete the matrix, we can use the following if statement: $IF(K_{call} > K_{put}, c - p, "")$ $e.g. = IF(I\$1 > \$H\$3, \$E2 - \$G\$3, "") =- 184.7$. This will provide the following output:

Strike price (Calls) / Strike price (Puts)	$2,480.00	$2,375.00	$2,265.00	$2,202.64	$2,157.41	$2,050.00	$1,940.00	$1,835.00
$ 2,480.00								
$ 2,375.00	-184.7							
$ 2,265.00	-109.38	-92.23						
$ 2,202.64	-74.63	-57.48	-25.87					
$ 2,157.41	-53.42	-36.27	-4.66	21.21				
$ 2,050.00	-16.57	0.58	32.19	58.06	80.82			
$ 1,940.00	3.92	21.07	52.68	78.55	101.31	168.88		
$ 1,835.00	12.35	29.5	61.11	86.98	109.74	177.31	263.76	

Here we can see that

1. **Combin1:** Highest net premium received $= 263.76$, which corresponds to $K_{call} = 1940 \; and \; K_{put} = 1835$
2. **Combin2:** Most neutral net premiums received $= 0.58$, which corresponds to $K_{call} = 2375 \; and \; K_{put} = 2050$
3. **Combin3**: Smallest net premium received (paid) $=- 184.7$, which corresponds to $K_{call} = 2480 \; and \; K_{put} = 2375$

To calculate the profit of each collar strategy given the spot price of Gold in 6 months, we can derive the following conditions based on the payoffs for a long put and short call, given that: $Option\ profit = payoff - premium$ and $Long\ Gold\ profit = S_0 - S_T$. Combining these two for ranges of K_{put} and K_{Call}, and using the fact that the payoffs for a long put and short call are $max(K - S_T, 0)$ and $-max(S_T - K, 0) = min(K - S_T, 0)$ respectively, we get:

When $S_T < K_{put}$	**When** $K_{put} < S_T < K_{Call}$	**When** $S_T \geq K_{Call}$
$K_{put} + c - p - S_0$	$(S_T - S_0) + c - p$	$K_{call} + (c - p) - S_0$

Putting these relations together using an $= IFS()$ statement in Excel, we can calculate the profit from the collar strategy for a given spot price of Gold in 6 months as, for the example of the $K_{call} = 1940$ and $K_{put} = 1835$ collar:

SUM

```
=IFS(A12<$F$9,$F$9+$E$8-$G$9-$B$7,AND($F$9<A12,$F$9<$D$8),(A12-$B$7)+$E$8-$G$9,A12>$D$8,$D$8+$E$8-$G$9-$B$7)
```

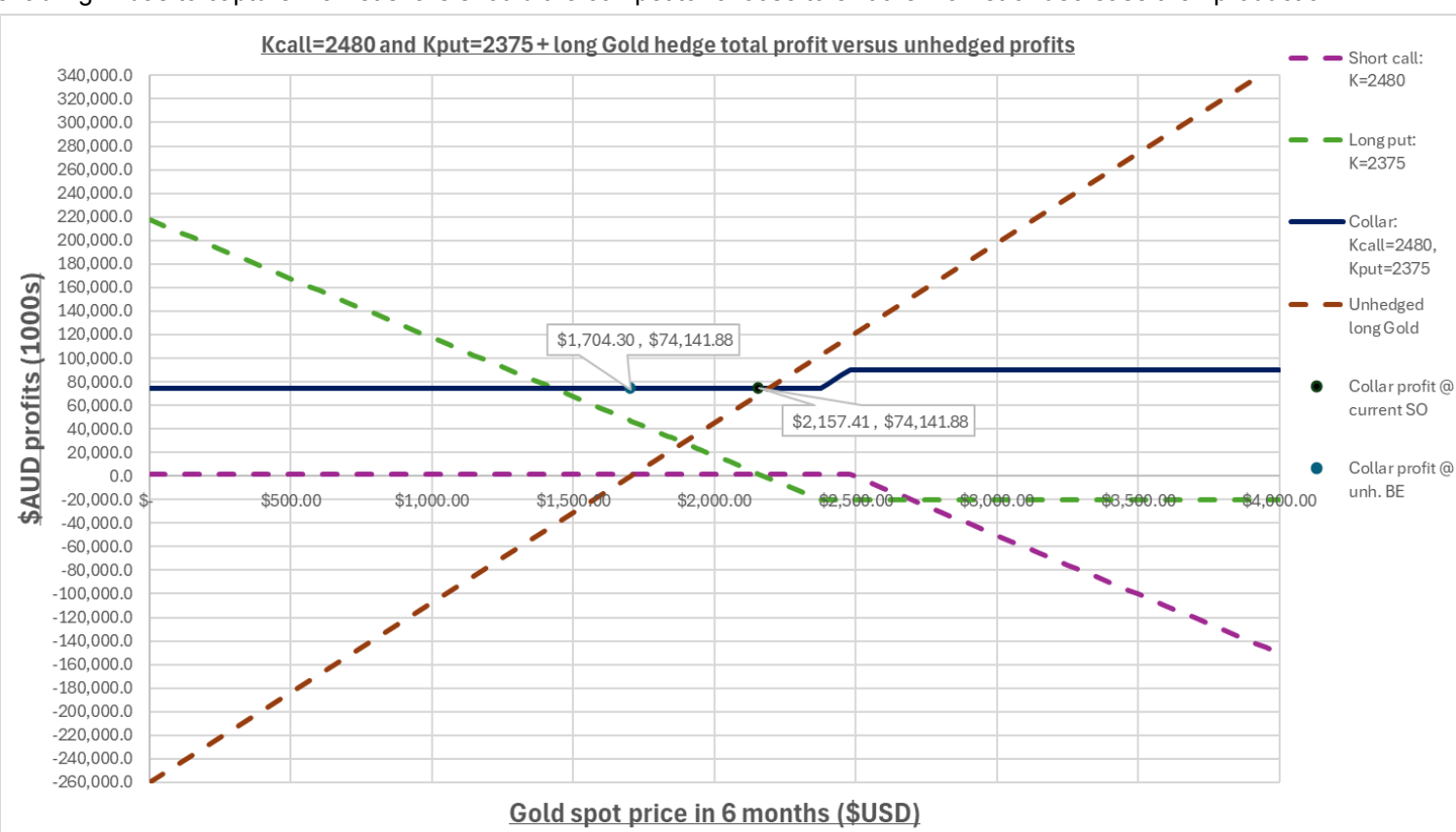
	A	B	C	D	E	F	G
1				Strike price (Calls) (USD per ounce)	OTC European call option premium (USD/ounce)	Strike price (Puts) (USD per ounce)	OTC Euro put option premium (USD/ounce)
2	Gold spot price(USD/ounce)	$ 2,157.41		$ 2,480.00	$ 15.94	$ 2,480.00	$ 285.57
3	Gold forward price for delivery in 6 months (USD per ounce)	$ 2,202.64		$ 2,375.00	$ 33.09	$ 2,375.00	$ 200.64
4	6-month US interest rate (c.c.)	5.65%		$ 2,265.00	$ 64.70	$ 2,265.00	$ 125.32
5	Exchange rate (XUSD/1AUD)	$ 0.6555		$ 2,202.64	$ 90.57	$ 2,202.64	$ 90.57
6	Costs(fixed per ounce) in AUD	2600		$ 2,157.41	$ 113.33	$ 2,157.41	$ 69.36
7	Costs USD	$ 1,704.30		$ 2,050.00	$ 180.90	$ 2,050.00	$ 32.51
8	q?=	1.50036%		$ 1,940.00	$ 267.35	$ 1,940.00	$ 12.02
9	Will sell (in 6 months)	100,000		$ 1,835.00	$ 360.99	$ 1,835.00	$ 3.59

Where $A12 = spot\ price\ of\ Gold\ in\ 6\ months\ time$. When we drag this down each row, changing only the spot price for each strategy combination, we get the profit for each strategy. Scaling this in another column by multiplying the profits by 100,000, we get the following total profit diagram for each collar strategy:

If financing the set-up of the strategy is constrained and will generate significant cash flow issues for Midas, then the $K_{call} = 2375\ and\ K_{put} = 2050$ collar is preferred. This is because it will provide a net cash flow at time zero of $\left(\frac{0.58\times100k}{0.6555}\right) = \$88,482.075\ (AUD)$, ensuring the hedging activities do not create short-term cash flow disruptions to Midas. This collar would also provide a minimum profit of $\$52.827 million\ (AUD)$ and a maximum profit of $\$102.407 million\ (AUD)$ depending on the spot price of Gold in 6 months, allowing Midas to participate in the benefits of Gold price appreciation and set a minimum on its sale price. This would be preferred over the $K_{call} = 1940\ and\ K_{put} = 1835$ collar, as although this provides the highest cash surplus at time zero, it significantly minimises the upside received from an appreciation in Gold prices in 6 months. This is substantiated by the consideration of the risk-to-return tradeoff defined by: $\frac{Average(hedged\ strategy\ profits - unhedged\ profits)}{premium}$ utilised in question c, which provides the following results:

Desc	Collar: K=1835 and K=1940	Collar: K=2050 and K=2375	Collar: K=2375 and K=2480
Average dif.	$ 22,198.15	$ 28,183.85	$ 33,317.65
Premium	$ 40,233.41	$ 88.48	-$ 28,176.96
Ratio	$ 0.55	$ 318.53	-$ 1.18

However, assuming there is no short-term cash flow risk to Midas, as the purpose of Midas undertaking hedging is not to make a profit but rather to focus on its core production activities and offset uncertainty in the spot price movement of Gold, the Collar, which minimises the downside risk should be preferred. Combination 3 ($K_{call} = 2480\ and\ K_{put} = 2375$) should therefore be chosen. As its maximum profit is also higher than combination 1, it will allow Midas to participate in the upside profits if the spot price of Gold increases over the next 6 months, which can help sustain its competitive advantage, as its profits would be similar to that of another producer with constant costs who do not hedge. It will also be able to do so without the additional risk of Combination 2's lower downside, traded for the potential of a greater upside profit than Combination 3 should the spot price of Gold increase over the next 6 months. Additionally, it will make Midas better off than a competitor with constant costs who does not hedge if the price of Gold decreases over the next 6 months, potentially enabling Midas to capture market share should the competitor choose to exit the market or decrease their production.



This would provide Midas with a minimum profit (lower bound) of $\$74,141,880(AUD)$ and a maximum profit (upper bound) of $\$90,160,000(AUD)$ depending on the spot price of Gold in 6 months. When $K_{2375} < S_T < K_{2480}$, Midas will also experience a profit exposure that is linearly related to the future spot price of Gold.

e) Compare the hedged profit from the collar strategy to the profit from a call bull spread strategy. What do you observe? Explain why. (5 marks)

The construction of a bull spread will require:

1. Long a call (C_1) with a strike price
2. Short a call (C_2) with strike price K_2
3. Ensure that $K_2 > K_1$, with both options having the same expiration date.

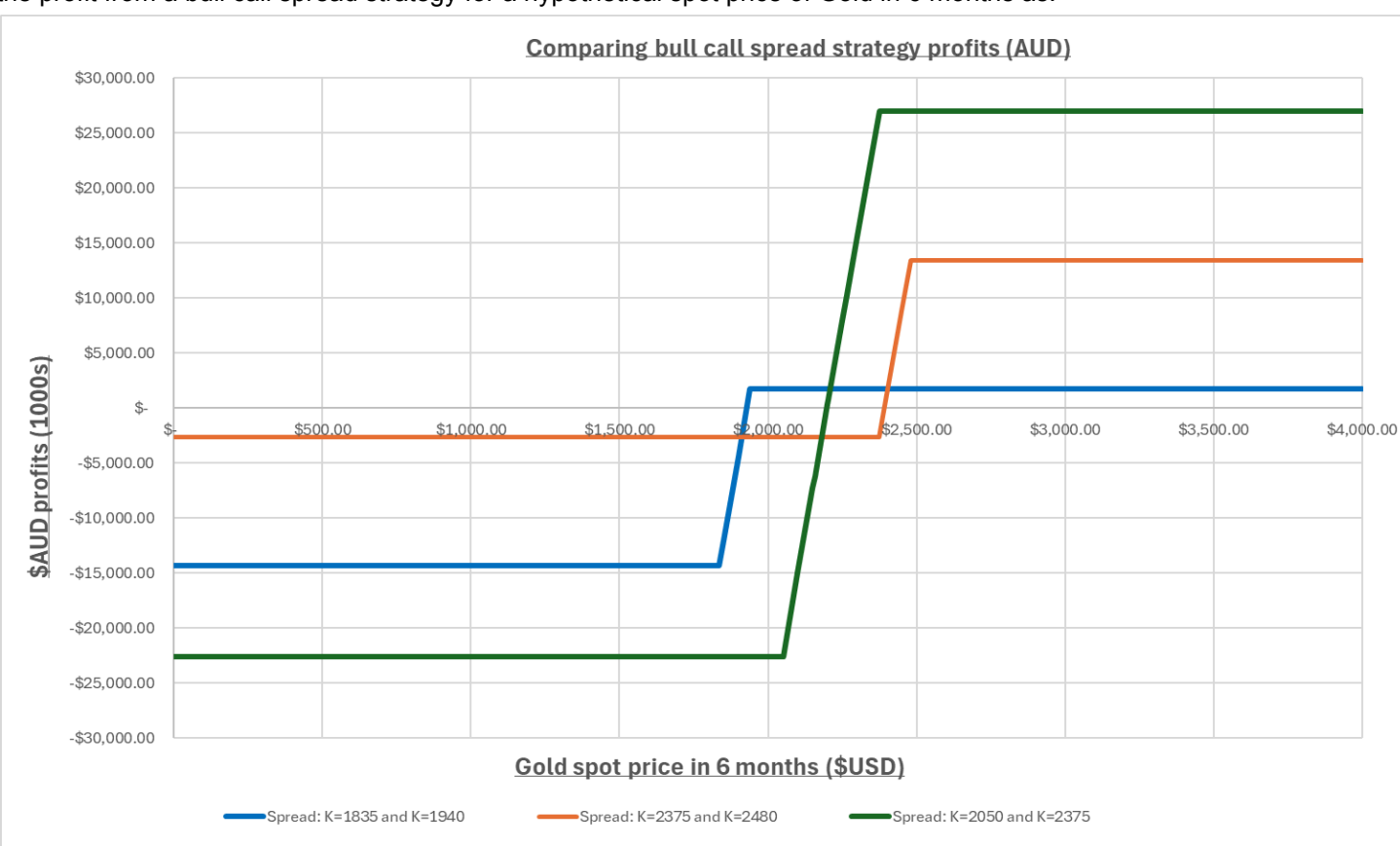
Unlike the collar, the bull call spread can be considered an income-generating activity to offset exposure rather than a pure hedge. This is because it cannot provide downside protection beyond the net premium paid/received. In the spread, the short call option limits hedged profits, and the long call option determines the strategy's downside. Calls with lower premiums will therefore reduce this net cost; however, they will be more OTM, thus offering less downside risk protection.

By ensuring that $K_{long\ call(bull\ spread)} = K_{long\ put(collar)} < K_{short\ call\ (bull\ spread)} = K_{short\ call(collar)}$ we can compare our 3 collar strategies with 3 bull call spreads that follow the same pairing. This will include replacing: $K_{call} = 1940\ and\ K_{put} = 1835$, $K_{call} = 2375\ and\ K_{put} = 2050$, and so on for each combination.

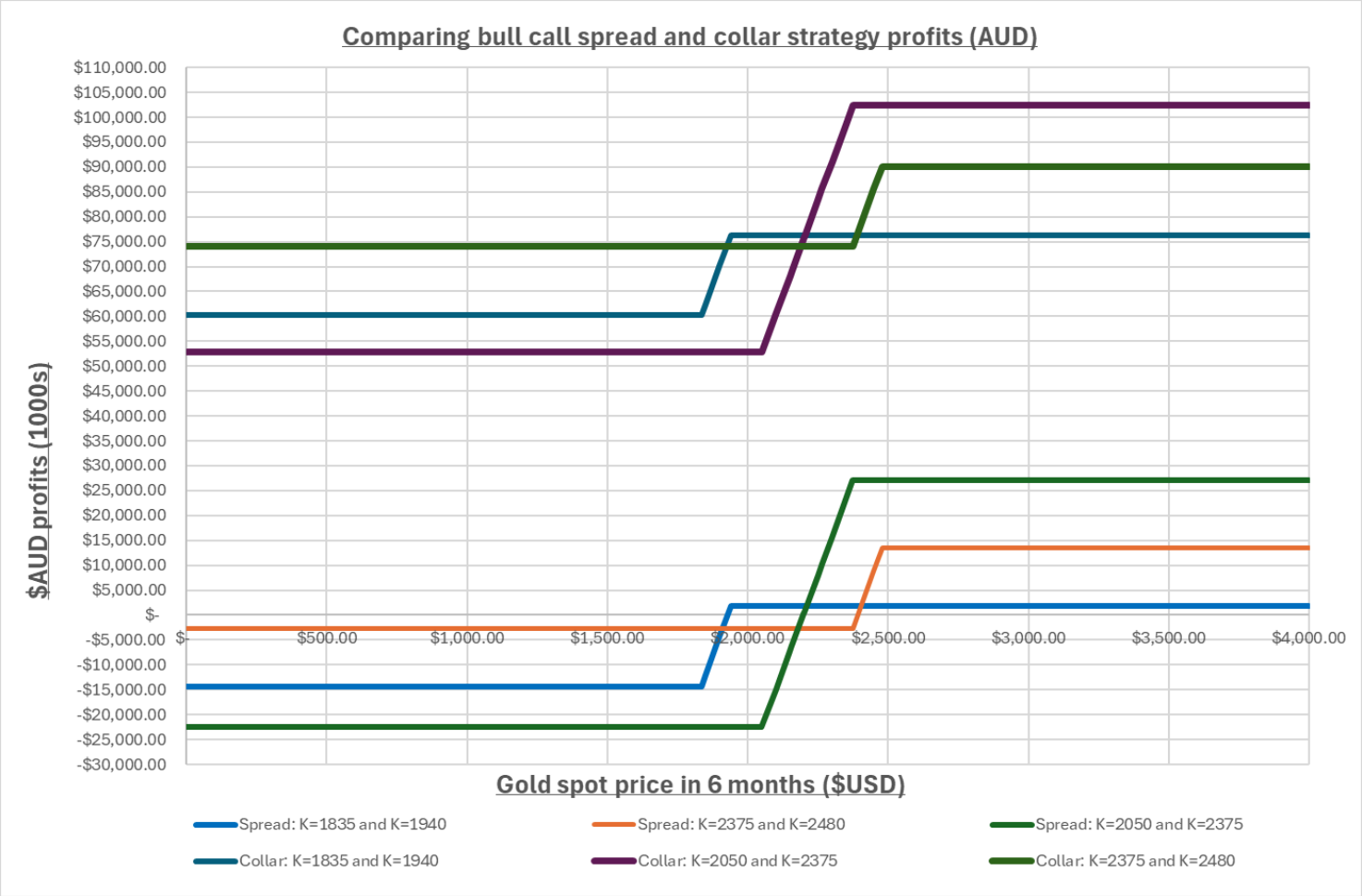
To calculate the profit of each call bull spread strategy given the spot price of Gold in 6 months and the associated profit diagrams, we can derive the following conditions based on the payoffs for a long call and short call: $Option\ profit\ =\ payoff\ -\ premium$. Noting also that the payoff from a long position in a call is: $max(S_T - K, 0)$ and the payoff from a short position in a call is $-\ max(S_T - K, 0) = min(K\ - S_T, 0)$, when combining these two for ranges of $K_{call\ short}\ and\ K_{call\ long}$, we get:

When $S_T \leq K_{Call\ long}$	**When** $K_{call\ long} < S_T < K_{call\ short}$	**When** $S_T \geq K_{call\ short}$
$(C_2 - C_1)$	$(S_T - K_1) + (C_2 - C_1)$	$(K_2 - K_1) + (C_2 - C_1)$

Putting these relations together using an $= IFS()$ statement in Excel, similar to that used for the Collar, we can calculate the profit from a bull call spread strategy for a hypothetical spot price of Gold in 6 months as:



The reason that the positions differ is derived from the fact that the bull call spread assumes only 2 positions: one in a long call and one in a short call. The collar strategy instead considers the underlying asset, Gold, which Midas already owns and does not need to buy at the market to have a long position. Instead, Midas incurs a fixed production cost of $2600AUD to procure one ounce of Gold. The collar, therefore, assumes 3 positions: a long position in Gold, a long put on Gold, and a short call on Gold. We can also compare the two diagrammatically:



For $S_T < K_1$: For the collar strategy, the long put is ITM and might be exercised, offering Midas downside protection with the net sale of Gold, and the short call (K2) expires OTM. For the bull call spread, there would be no transactions in the underlying asset (Gold), with both calls expiring worthless, resulting in a loss limited to the premium paid.

For $K_1 < S_T < K_2$: In the collar, both the long put and short call expire OTM, maintaining the long position in Gold and ensuring the premium on the short is kept. For the bull call spread, the long call becomes ITM, increasing in value as ST approaches K2, while the short call stays OTM, resulting in no net transactions in Gold.

For $S_T \geq K_2$: For the collar, if ST exceeds K2, the put expires OTM, and the short call would be exercised by the counterparty, leading to the sale of Gold at K2, which is where the upper bound on this collar profit function is located. In the bull call spread, both options are ITM, leading to neutral Gold transactions if both are exercised. This also characterises the upper bound of the spread strategy, maximising the profit to the spread distance plus the upfront premium. Such a position would have to be offset in the Gold spot market, neutralising any net purchase or sale of Gold for Midas beyond what it independently chooses to produce.

We can therefore see that each collar and spread pair have the same turning points and lower and upper bound ranges for a given spot price of Gold in 6 months. Overall, the collar uses a long put to set the minimum sale price, capping downside risk exposure, and a short call to offset the premium costs, limiting the maximum profit. Contrarily the spread's maximum profit is capped at the difference between the two strike prices plus the net premium, with the lower bound representing the net premium when both calls expire worthless. In aggregate, these instruments result in different magnitudes of profits despite having similar derivatives in the collar and spread profit functions for a given spot price of Gold.

Question 3: Recommendation (30 marks)

The analysis above highlights the following strategies for Sons of Midas: unhedged, hedged with a forward, hedged with a put option, or hedged with a collar. Which strategy would you recommend to Sons of Midas? Your answer should clearly state the arguments for and against each strategy in relation to Sons of Midas (which would include a discussion of the risk-return tradeoffs), followed by a recommendation and conclusion.

Introduction

The decision to hedge is fundamentally influenced by Midas's capital structure and organisational goals. While hedging could potentially reduce Midas's profit volatility, a key concern for the CFO, it also entails significant upfront costs. The 0.5 debt-to-equity ratio further indicates that Midas is not fully leveraging the advantages of financing its operations. Hence, any hedging strategy must balance the expectations of Midas's debt and equity holders, as lenders typically favour low volatility in profits to reduce the risk of defaults, while equity holders demand an equity risk premium (return) on their investments.

Unhedged strategy

The substantial costs of puts and collars needed for hedging can significantly disrupt Midas's cash flows, totalling $\$43.5652 million\ (AUD)$ for the K=2480 put. Instead, the unhedged strategy would provide Midas full exposure to the prevailing Gold price in 6 months without any upfront costs. However, this does come with the tradeoff of the smallest possible lower bound in profits among all strategies. Given that Gold's annualised volatility over the last 30 years was 15.44% (Oxford Gold Group, 2023), the 6 monthly volatility is less than that which would result in Midas breaking even on its unhedged production and sale of Gold: $2157.41 \times (1 - (0.1544 \times \sqrt{0.5}) = 10.92\%.\ ) = 1921.82\ USD$ (68% of the time assuming a normal distribution). However, if Midas does not hedge and Gold prices decline, this creates liquidity risks, signalling to shareholders and lenders that Midas is undertaking unnecessary risks. Conversely, if Gold prices rise, Midas could outperform hedged competitors. However, to maintain shareholders' confidence and meet lender requirements, Midas must implement a hedging strategy to mitigate Gold price uncertainty and enhance its creditworthiness. This approach is supported by common banking practices that require mining companies to have hedging strategies in loan agreements to mitigate financial risks (Babenko et al. 2024).

Hedging with a short forward

Contrarily, utilising the 6-month short forward contract at a price of $\$2,20.64 USD$ would enable Midas to secure a sale price for Gold in six months without any upfront costs. This would reduce the uncertainty of its future cash flows, ensuring Midas meets its short-term debt obligations regardless of the price of Gold. However, the OTC market does carry some credit risk, as the counterparty of the forward contract could default on its obligations, resulting in Midas needing to find new buyers, likely at lower prices. Additionally, the short forward contract would limit Midas's exposure if Gold rises above the strike price, as it provides a constant bound on profits. This opportunity cost is captured by $F_0 - S_T$, which is negative when $S_T > 2202.64$. Furthermore, Midas's Gold output could fluctuate due to operational issues or unexpected shocks. As forwards are highly customised, specific delivery timelines could create discrepancies between the forward contract and actual production, disrupting Midas's cashflows.

Hedging with a put

This strategy would allow Midas to sell Gold at a predetermined strike price, providing a minimum bound on profits while allowing the firm to benefit from Gold price increases. This overcomes the upper bound limitations of the short forward and poses less credit risk, as the option can be purchased on an exchange (generally requiring margin) rather than OTC. However, unlike forwards, the use of a singular put option with a strike price $K = \$2480(USD)$ is more expensive than all other forms of hedging, at a substantial total upfront cost of: $\$43.5652 million\ (AUD)$. This could create significant cash flow problems for Midas, even if it opted to use a cheaper strike that still ensures positive profits after 6 months, such as the 1835 at $3.59USD per ounce.

Hedging with a collar

A collar using a short call at $K = 2375$ and long put at $K = 2050$ reduces the cost of the put, paying less premium $\left(\left(\frac{0.58 \times 100k}{0.6555}\right) = \$88,482.075\ (AUD)\right)$ whilst providing a lower bound profit of $\$52.827 million\ (AUD)$ and a maximum of $\$102.407 million\ (AUD)$. This ensures Midas has access to downside protection but at a lower cost. For increases greater than the short call strike, Midas's profits will be limited to $\$102,407,322.65\ (AUD)$ as the call option sets an upper bound

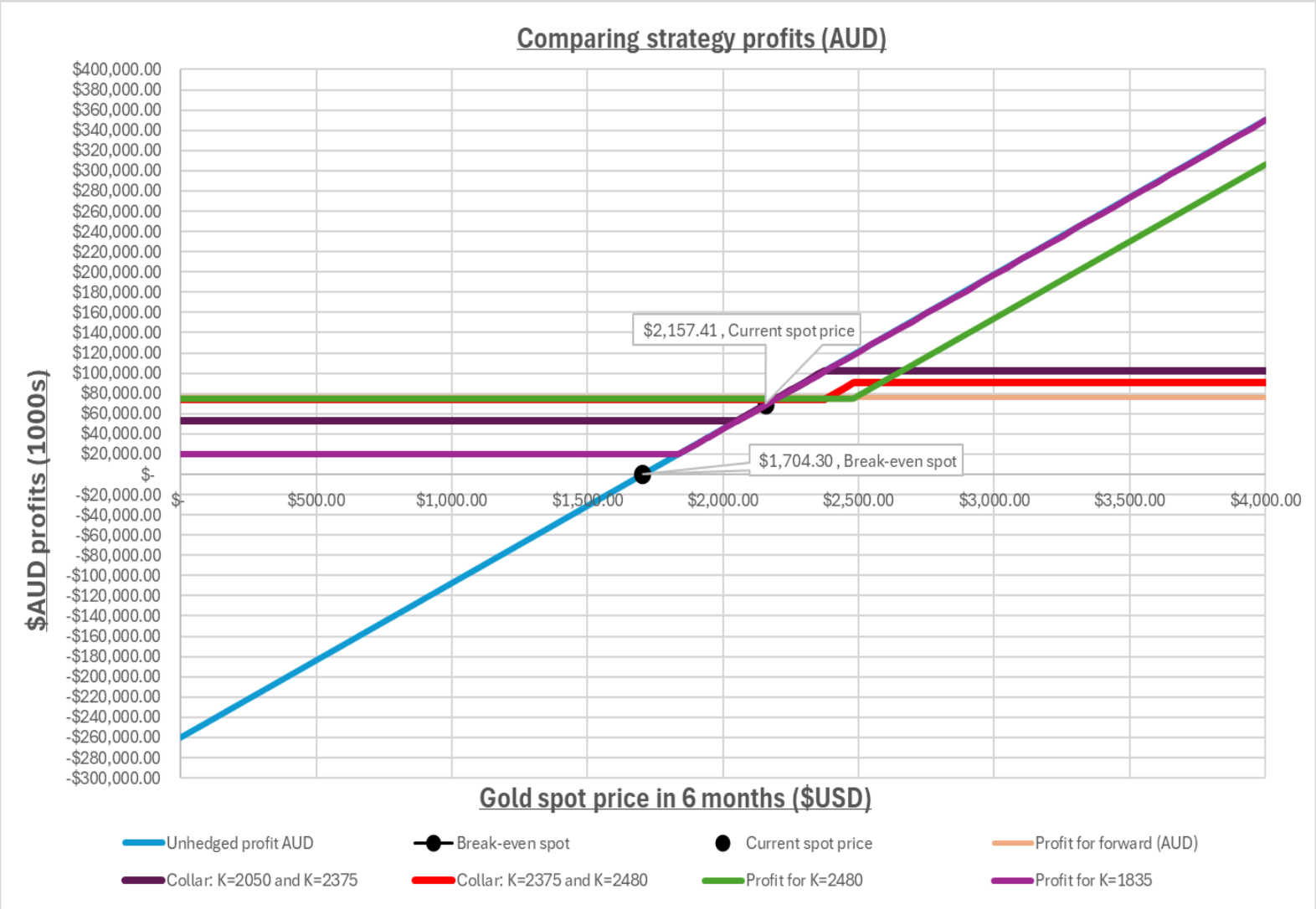
on the sale price, limiting the exposure to volatile Gold prices. In this way, for both the collar and the put, if the Gold price does not fall to the put strike price level, the put's cost would have been unnecessary, lowering profits below that received if otherwise unhedged.

Recommendation

In order to address the expectations of its debt and equity holders, Midas must hedge its Gold sale price risk using the $K = 2375$ $K = 2050$ collar. This will provide a net cash inflow at the time of the hedge, circumventing short-term liquidity issues whilst allowing Midas to focus on its core production activities. This collar's linear payoff between its strikes (when $K_{put} < S_T < K_{Call}$) will also provide a lower turning point in its profit function, allowing Midas to experience more exposure to the potential increases in the price of Gold than other hedging alternatives without substantial upfront costs. Assuming no risk of default on the options, this defined range of profitability will reduce uncertainty whilst enabling Midas to obtain a competitive advantage over firms that do not hedge their price risks.

Conclusion

Given the risk-reward tradeoffs, remaining unhedged, or entering a short forward contract, are not optimal strategies for Sons of Midas. While the forward provides stability in expected profits, it lacks flexibility by limiting upside potential and bearing credit risk. The choice to use the collar strategy over the put will ensure that the expectations of both equity and debt holders are met. The lower and upper bounds will provide a more stable outlook on future cash flows for Midas, reducing the uncertainty of its financial planning activities, allowing Midas to access favourable credit terms for expansion activities in other regions. Overall, each feasible strategy available to Midas can be summarised diagrammatically:



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